MATH 201 Your name: _____

**Chapter 2 checkpoint!**

Chapter 1 scorecard:

| Learning target: | DF1 | DF2 | DFa | DFb | AD2 |
|---|---|---|---|---|---|
| Your confidence level before starting (0-5): | | | | | |
| Your confidence level after the quiz (0-5): | | | | | |
| The mark you earned on this attempt: | Success! Try again! | Success! Try again! | Success! Try again! | Success! Try again! | Success! Try again! |

Chapter 2 scorecard:

| Learning target: | 2.1 | 2.3 | 2.5 | 2.7 | 2.all |
|---|---|---|---|---|---|
| Your confidence level before starting (0-5): | | | | | |
| Your confidence level after the quiz (0-5): | | | | | |
| The mark you earned on this attempt: | Success! Try again! | Success! Try again! | Success! Try again! | Success! Try again! | Success! Try again! |

Before anything else, please do the following:

- Rank your confidence from 0-5 on each of the learning targets. 5 means "I could teach a whole class about this;" 0 means "I am genuinely not sure I have heard these words before."
- Write your name on this page and on each of the other pages of the quiz.

Then do the quiz! Some reminders:

- Open notes, closed computer.
- If you need more room to write, use the back of the same learning target page, or ask me for some scratch paper.
- Read the questions carefully and make sure you're answering each part.
- Use good grammar for derivative problems.
- Show all your work and explain all your thinking!

When you are done:

- Rank your confidence from 0-5 on each of the learning targets. 5 means "I absolutely nailed that question for sure;" 0 means "oof, I definitely didn't get that one."
- Make double sure your name is on every page, including any scratch paper.
- Hand in your work, separated by learning target.

Have fun and do your best! I believe in u ♡

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**Learning target 2.1, version 1**

Compute derivatives using basic function rules, the constant multiple rule, and the sum rule.

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Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $f(x) = 2\cos(x) + 4e^x$

2. $h(t) = \sqrt[3]{t^4} + \dfrac{4}{t^4}$ (Don't use the quotient rule or the chain rule.)

3. $g(w) = 6w^3 - 2w^2 - 9w - 2$

4. $C(t) = 64(e^7 + 7^e) - \dfrac{\sqrt[27]{193262}}{16 - \pi^2} \cdot \sin(4)$ (Hint: Look carefully!)

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**Learning target 2.3, version 1**

Compute derivatives using the product and quotient rules.

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1. Consider the function $g(t) = 7 \cdot t^4$.

   (a) Compute $g'(t)$ using the product rule. Simplify as much as possible.

   (b) Now compute $g'(t)$ again, *without* the product rule.

   (c) Are both ways *correct*? Which way do you *prefer*?

2. Consider the function $f(t) = \dfrac{t^4}{7}$.

   (a) Compute $f'(t)$ using the quotient rule. Simplify as much as possible.

   (b) Now compute $f'(t)$ again, *without* the quotient rule.
   (Hint: What's another way to say "divide by 7"?)

   (c) Are both ways *correct*? Which way do you *prefer*?

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**Learning target 2.5, version 1**

Compute derivatives using the chain rule.

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Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $h(t) = 16\left(t + e^t + 2\right)^4$

2. $f(w) = 5\sin\left(w^{7/3}\right)$

3. $g(w) = 5\left(\sin(w)\right)^{7/3}$

4. $p(y) = 2\ln\left(\sin(y) + y^2 + 3\right)$

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**Learning target 2.all, version 1**

Compute derivatives of complicated functions using a combination of derivative rules.

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Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $g(x) = \sqrt{\cos(-2x^4 + 5)}$

2. $f(w) = \left(\dfrac{6w^6 - 1}{5w^2 + 3}\right)^4$

3. $h(y) = \left(3y^2 + 7y\right)^4 \cdot y^{1/4}$

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**Learning target 2.7, version 1**

Compute derivatives of implicitly-defined functions.

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Consider the curve defined implicitly by the equation

$$y^3 - x \cdot y = -6.$$

1. Use implicit differentiation to find an expression for $\dfrac{dy}{dx}$ that depends on $x$ and $y$. Show all of your work and thinking clearly, using correct notation.

2. The point $(7,2)$ lies on this curve. (I promise this is true, but how would you check?)
   Find the exact slope of the tangent line to this curve at the point $(7,2)$. Show clearly how you determined this value, and label your work with proper notation.

3. (Also LT 1.8!) Figure out the equation of the tangent line $L(x)$ to this curve at the point $(7,2)$, and use your tangent line equation to make a reasonable guess about the $y$-coordinate of the point $(6.8, ____)$.